

CFD Investigation: Shear-Driven Open Cavity Flow

Parametric Reynolds Number Analysis ($Re = 100 - 3000$) via ANSYS Fluent

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Discipline: Mechanical Engineering (Fluid Dynamics / CFD)

Solver: ANSYS Fluent (Coupled / Second-Order Upwind)

Domain: Channel Flow over Recessed Cavity

Cavity Dimensions: $1\text{ cm} \times 1\text{ cm}$ ($W/D = 1.0$)

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1. Overview & Flow Physics

Flow over an open rectangular cavity represents a canonical shear-driven recirculation problem encountered in aerospace configurations (landing gear wells, internal bomb bays) and industrial heat exchangers. As the channel boundary layer passes over the recessed slot ($1\text{ cm} \times 1\text{ cm}$), a detached free shear layer forms across the cavity mouth. Momentum transfer through viscous and inertial stresses drives the trapped fluid into an enclosed recirculation zone.

This study tracks the evolution of internal vorticity, shear layer penetration depth, and vortex core migration across seven Reynolds numbers: $Re = 100, 200, 400, 600, 800, 1000$, and 3000 .

2. Parametric Vorticity & Recirculation Regimes

Case	Reynolds No. (Re)	Vortex Core Location	Internal Recirculation Dynamics
1	100	Cavity geometric center	Weak, diffusion-dominated viscous recirculation
2	200	Near center	Stable, well-formed circular primary vortex
3	400	Shifting downstream	Growing convective entrainment and circulation
4	600	Mid-right region	Transition to inertia-dominated internal rotation
5	800	Downstream shifted	High boundary shear and increased core vorticity
6	1000	Lower-right quadrant	Deep shear layer dip, intensified cavity swirl
7	3000	Compressed on right wall	Strong downstream wall impingement & peak vorticity

Table 1: Evolution of cavity vortex dynamics and spatial migration across Reynolds numbers.

3. Hydrodynamic Analysis: Reynolds Number vs. Vorticity

- **Vorticity Intensification with Reynolds Number:** As Re increases from 100 to 3000, convective momentum transfer across the cavity opening strengthens significantly. The free shear layer supplies continuous angular momentum into the cavity, causing the internal circulation velocity and local vorticity ($\omega = \nabla \times \vec{V}$) to rise monotonically with Re .
- **Vortex Core Displacement:** At creeping/low Reynolds numbers ($Re = 100 - 200$), the recirculating flow is symmetric and viscous-dominated, placing the vortex core near the geometric centroid of the cavity. As inertial forces dominate with rising Re , the core is progressively driven toward the downstream (right) vertical wall and shifted deeper into the lower-right quadrant.
- **Shear Layer Deflection and Impingement:** At higher Reynolds numbers ($Re \geq 1000$), the shear layer plunges deeper into the cavity mouth before impinging directly onto the downstream upper corner, steepening velocity gradients along the cavity walls.

4. Velocity Distribution & Vortex Evolution ($Re = 100$ to $Re = 3000$)

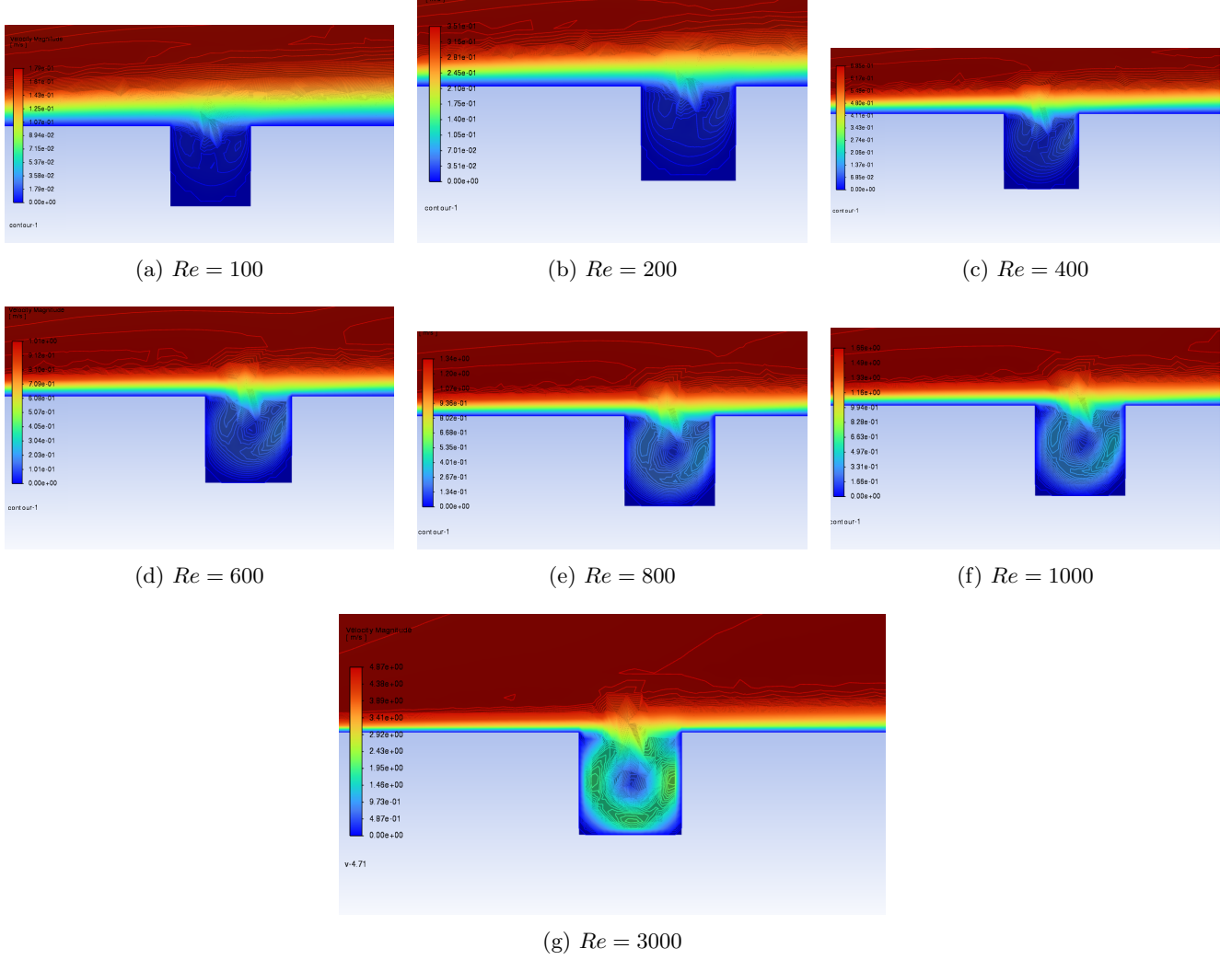


Figure 1: Velocity contours illustrating the progressive increase in internal cavity vorticity, shear layer deflection, and downstream vortex core migration as Re increases.

5. Summary & Conclusion

- Across the investigated range of $Re = 100$ to $Re = 3000$, the cavity flow transitions from a symmetric, viscous-dominated diffusion loop to an inertia-driven high-vorticity vortex.
- **Primary Finding:** Increasing the Reynolds number directly enhances the circulation strength and vorticity within the cavity, while shifting the vortex core downstream toward the impingement wall.
- High-resolution second-order spatial discretization in ANSYS Fluent captured the shear layer deflection without numerical instability across all cases.